

Renewable energy in the EU

Saša Butorac and Agnieszka Widuto, Members' Research Service; Graphics: Giulio Sabbati

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Summary

Europe's key instrument to achieving energy independence and increasing competitiveness lies in the energy transition and, specifically, in boosting the generation capacity of renewable sources of energy. Following the European Green Deal and 'fit for 55' initiatives, the EU legislative framework for achieving this is largely in place. Significant progress has been made, in particular since the launch of the REPowerEU initiative in May 2022 in the wake of Russia's full-scale invasion of Ukraine. Member States have increased the share of renewables in their energy mix, and the EU is consistently progressing towards its target of a 42.5 % share of renewables in final energy consumption by 2030. The share of renewables in sectors such as electricity (47.5 % of final energy consumption in this sector), heating and cooling (26.7 %) and transport (11.2 %) is also increasing, although progress has been fastest in terms of electricity. The main challenges to an accelerated deployment of renewables can be identified as the cost of capital, timely development of the grids, and the complex and lengthy permitting procedures both at European and national level.

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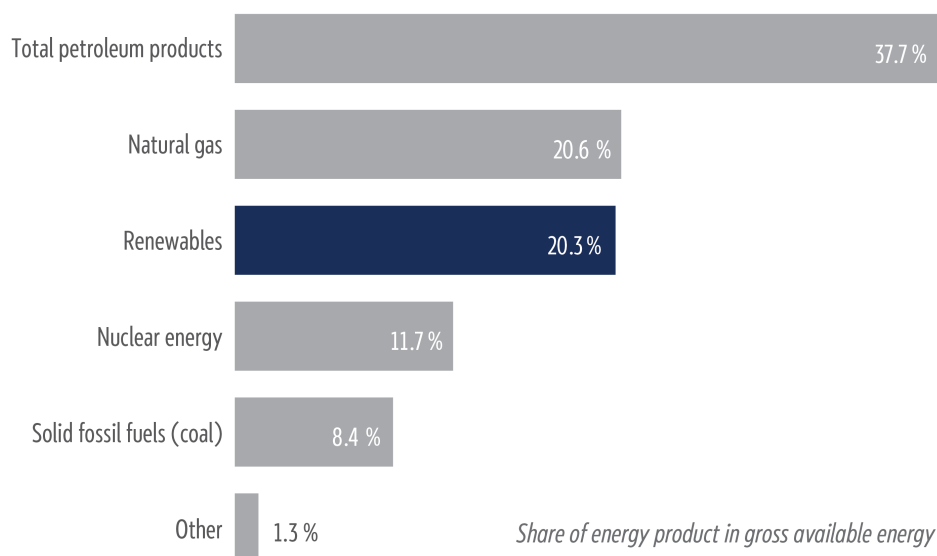


Introduction

Boosting renewables is a key element of the EU's energy transition. The use of renewable energy helps reduce greenhouse gas (GHG) emissions and increases energy security by diversifying energy supplies and reducing dependency on fossil fuels, which currently still dominate the EU energy mix (see Figure 1). Oil, gas and coal add up to about 67 % of gross available energy in the EU.¹ Renewables account for about 20 %, followed by nuclear power, which makes up almost 12 % of the EU energy

mix. According to the Centre for Research on Energy and Clean Air, the EU's fossil fuel imports in 2025 amounted to approximately €396 billion, versus [around €330 billion](#) spent on the energy transition.

Figure 1 - EU energy mix, 2024

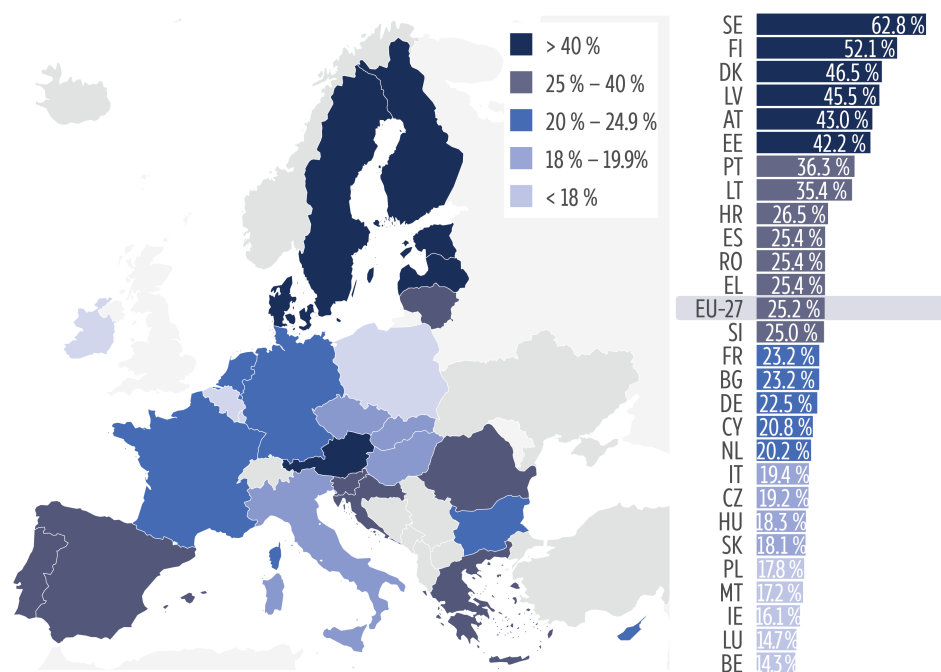


Data source: [Eurostat](#), 2026.

In 2024, renewable energy accounted for [25.2 %](#) of final energy consumption in the EU. ² This means that the EU will still have to make significant efforts to reach the 42.5 % target set in the [Renewable Energy Directive](#) in order to bridge the gap of over 17 percentage points (pp).

The share of renewable energy in final energy consumption varies widely across Member States (see Figure 2), with Sweden achieving the highest share in 2024 (63 %), followed by Finland (52 %), and Belgium and Luxembourg the lowest shares (below 15 %). The European Commission's latest EU-wide assessment of the [national energy and climate plans](#) (NECPs), published in May 2025, estimates that Member States are almost on track to achieve the collective EU target by 2030, with a 1.5 pp gap remaining. To assess this, the Commission first used a special calculation formula to estimate the national contributions – expressed as the share of renewables in each country's consumption – necessary to achieve the overall EU target. It then compared them to the renewables share that is projected to be achieved by 2030 on the basis of national policy frameworks outlined in each NECP (see Table 3 in the [annex](#) to the assessment).

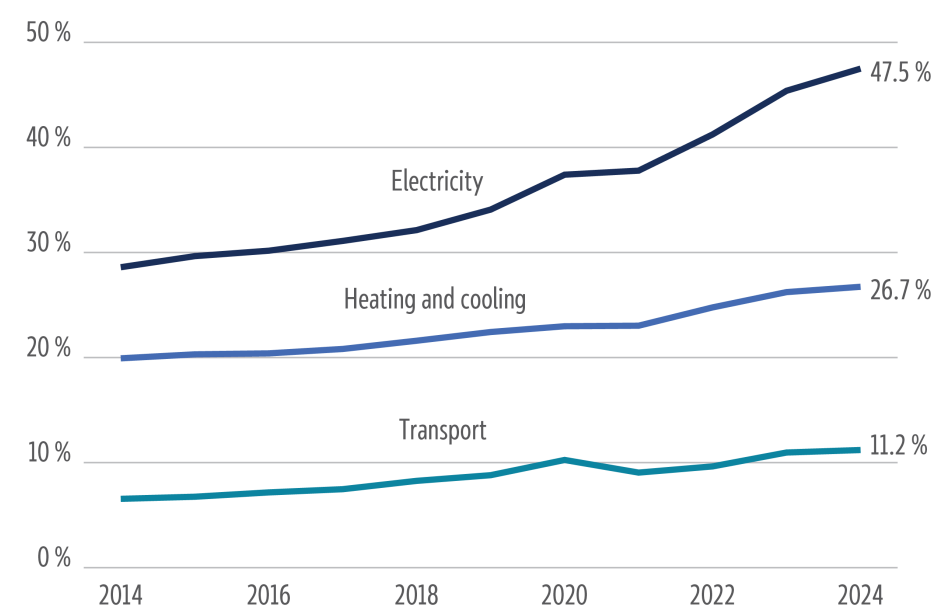
Figure 2 - Share of energy from renewable sources in final energy consumption, 2024



Data source: [Eurostat](#), 2026.

According to the 2025 [State of the Energy Union](#) report, the overall share of renewable energy has been increasing by 0.8 pp annually since 2020. Progress has been particularly strong in the electricity sector, with an increase in the renewables share in final energy consumption from 37.4 % in 2020 to 47.5 % in 2024. Progress has been limited in both heating and cooling (from 23 % in 2020 to 26.7 % in 2024) and in transport (from 10.2 % in 2020 to 11.2 % in 2024). The Renewable Energy Directive sets mandatory national targets for heating and cooling (0.8 % annual increase in the share of renewable energy sources until 2026, followed by a 1.1 % annual increase until 2030) and transport (Member States can choose between a 14.5 % reduction in GHG intensity or ensuring a renewables share of at least 29 % by 2030), without any specific percentage targets for renewables in electricity. However, the [Clean Industrial Deal](#) includes the key performance indicator to annually install 100 gigawatts (GW) of renewable electricity capacity up to 2030.

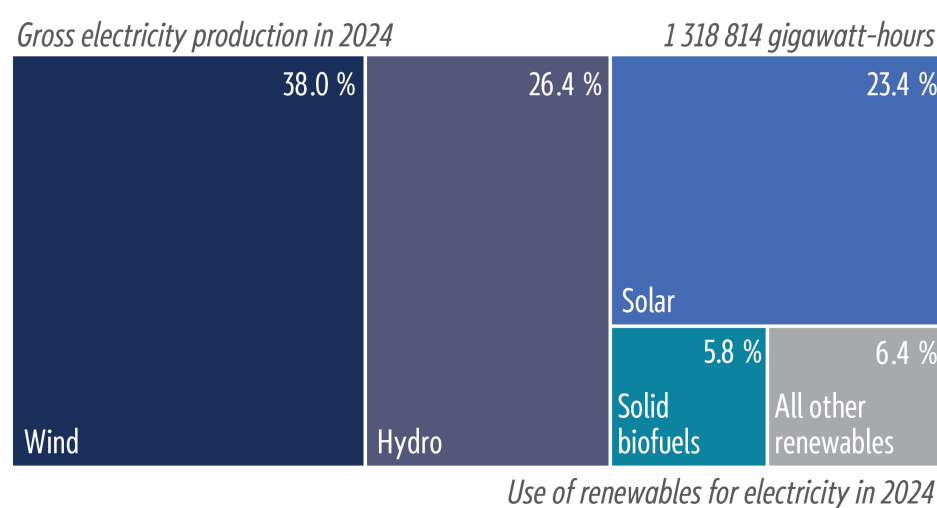
Figure 3 - Share of energy from renewable sources in final energy consumption, 2014 to 2024



Data source: [Eurostat](#), 2026.

At the EU level, in 2024, wind had the highest [share](#) among renewables in electricity generation (38 %), followed by hydropower (26.4 %) and solar (23.4 %). Solid [biofuels](#) accounted for only 5.8 %, while other renewable sources made up the remaining 6.4 % (see Figure 3). Solar power is the fastest-growing source: in 2020, it accounted for [5.2 %](#), and has grown more than fourfold since.

Figure 4 - Share of renewable energy sources in EU electricity production, 2024



Data source: [Eurostat](#), 2026.

Legislative framework

The **Renewable Energy Directive** (RED, [Directive 2018/2001/EU](#)) is the key EU legislative act setting the policy framework for the development of renewable energy sources in the EU. Following adoption of the [European Green Deal](#) in 2019, a revision of the directive was started in July 2021 (within the 'fit for 55' initiative), with the aim of boosting the deployment of renewables to reach the EU 2030 climate target (a reduction of 55 % of EU GHG emissions compared with the 1990 baseline). In the wake of

Russia's full-scale invasion of Ukraine in February 2022 and the subsequent launch of [REPowerEU](#), the Commission proposed a series of measures to further accelerate renewable energy projects; these specifically targeted the permit-granting procedures included in the revised directive, which was adopted in October 2023. The Commission came forward with additional measures designed to facilitate permitting procedures in its European grids package [proposal](#) published in December 2025.

The RED sets an EU-wide binding target to increase the share of renewables to at least 42.5 % of the EU's gross final energy consumption by 2030; binding targets for national contributions to the EU-wide target are not explicitly laid down. The directive is a key element of the overall EU energy framework that accelerates the transition away from fossil fuels, cuts GHG emissions, and increases the security of energy supply in the EU. It also promotes sector-specific goals in the heating and cooling sector, in transport and in industry.

Key aspects of the RED include:

- **binding 2030 targets:** a legally binding EU-level goal to achieve a 42.5 % share of renewables in energy consumption by 2030, with an additional non-binding 2.5 % top-up for a 45 % target;
- **sectoral targets:** specific targets for introducing renewable energy into industries, buildings, heating and cooling, and transport (see below), to speed up adoption in harder-to-decarbonise areas;
- **faster permitting:** simplified and shortened permitting procedures for renewable energy projects, particularly for solar and wind power, to facilitate their deployment;
- **consumer and community empowerment:** provisions that allow citizens to engage in renewable energy projects, including self-consumption and forming renewable energy communities;
- **sustainability criteria:** strengthened criteria for biomass and other bioenergy to ensure they are environmentally sustainable, reducing the impact on biodiversity;
- **transposition:** Member States were supposed to transpose the directive by 21 May 2025, with an earlier deadline for permitting rules (1 July 2024); in July 2025, 26 Member States (all except Denmark) received formal [notices](#) from the Commission for failing to meet the May 2025 deadline.

The RED comprises the following sub-targets.

- **Industry:** the RED sets binding targets for renewable fuels of non-biological origin (RFNBOs), requiring 42 % of hydrogen used in industrial processes to be renewable by 2030 (60 % by 2035). It also mandates an annual 1.6 pp increase in renewable energy use by the industry. Member States may reduce the 42 % RFNBO target by up to 20 % if they meet their national contribution to the overall EU target, and if their share of hydrogen from fossil fuels does not exceed 23 % in 2030 (and 20 % in 2035). RFNBOs are gaseous or liquid fuels from non-biomass renewable sources (e.g. green hydrogen, synthetic fuels). They must achieve at least 70 % GHG emissions savings. A [delegated act](#) on RFNBOs, based on the RED, mandates strict additionality criteria to ensure production uses new, renewable electricity.
- **Transport:** the RED sets a binding 2030 target for the transport sector in Member States to achieve either a 14.5 % reduction in GHG intensity or a 29 % share of renewable energy. It also introduces a 5.5 % combined share for advanced biofuels and RFNBOs in transport, with a specific 1 % minimum for RFNBOs.

- **Heating and cooling:** the RED mandates an increased share of renewables in heating and cooling, requiring Member States to raise this share by at least 0.8 pp annually until 2025 (as an average for 2021 to 2025) , and by 1.1 pp from 2026 to 2030. It sets an indicative 49 % renewable energy target for **buildings** by 2030, and promotes waste heat recovery and district heating, aiming to phase out fossil fuel boilers. The target for renewable energy in district heating and cooling increases to an average annual rate of 2.2 %. Member States must comply with mandatory annual increases in renewable energy in the sector, moving away from previous voluntary targets.

Legislative proposal on acceleration of permit-granting procedures

The Commission identified permit-granting procedures as one of the key hurdles towards faster deployment of renewable sources of energy. To address this challenge, it put forward a **legislative proposal** on acceleration of permit-granting procedures on 10 December 2025 as part of the [European grids package](#).

Specifically, the Commission identified incoherent administrative systems, inadequate staffing in national authorities, the duration of environmental assessments, lack of public acceptance, limited digitalisation and data availability, as well as administrative and judicial challenges as the main causes of slow permitting procedures. The proposal aims to address these bottlenecks by:

- **reducing deadlines** and avoiding unnecessary delays in the permitting process by, among other things, including relevant [provisions](#) from [Council Regulation \(EU\) 2022/2577](#) on accelerating the deployment of renewable energy that were not integrated into [Directive \(EU\) 2023/2413](#) (the Revised Renewable Energy Directive – RED III);
- introducing additional **flexibility in the application of environmental rules**;
- introducing a **permitting regime for electricity grids** at EU level that is based on the approach to the permitting of gas and hydrogen assets in [Directive \(EU\) 2024/1788](#) (the Gas Market Directive) and aligns with the framework of the RED and Regulation (EU) 2022/869 on guidelines for trans-European energy infrastructure (the [TEN-E Regulation](#)).

The European Parliament started its work on the [proposal](#) in January 2026. The Committee on Industry, Research and Energy (ITRE) oversees Parliament's work on the [file](#).

Main renewable energy sources

According to the [definition](#) in the Renewable Energy Directive, 'energy from renewable sources' or 'renewable energy' means 'energy from renewable non-fossil sources, namely wind, solar (solar thermal and solar photovoltaic) and geothermal energy, osmotic energy, ambient energy, tide, wave and other ocean energy, hydropower, biomass, landfill gas, sewage treatment plant gas, and biogas'. Some of these energy sources have binding or indicative targets in EU legislation and strategies (see Table 1).

Table 1 - State of play and EU 2030 targets for selected renewable technologies

	2024-2025	EU 2030 target	Legal basis/EU source
Solar	406 gigawatts (GW)	Additional solar photovoltaic capacity of almost 600 GW (total of 750 GW)	Solar energy strategy (2022)
Wind	21 GW offshore wind	60 GW of offshore wind	Offshore renewable energy strategy (2020)
Renewable hydrogen	n/a	10 million tonnes (production) 10 million tonnes (imports)	REPowerEU (2022)
Heat pumps	21.5 million	41.5 million	REPowerEU (2022)
Electrolysers	308 megawatts (MW)	40 GW	EU hydrogen strategy (2020)
Biomethane	n/a	35 billion cubic metres (bcm)	REPowerEU (2022)
Biogas	n/a	12.3 million tonnes of oil equivalent (Mtoe)	REPowerEU (2022)

Data source: EPRS, based on relevant EU documents for the 2030 targets and the following sources for 2024-2025: Solar Power Europe [data](#), 2025; Wind Europe [data](#), 2025; JRC Clean Energy Technology Observatory [data](#) on heat pumps, 2024; ACER [data](#) on renewable hydrogen electrolysers, 2024.

Solar energy (photovoltaic and thermal)

According to a 2025 Solar Power Europe [report](#), the EU reached a solar photovoltaic (PV) capacity of 406 GW in 2025. The EU [solar energy strategy](#) of 2022 set a non-binding target for additional solar PV capacity of 320 GW by 2025 and almost 600 GW by 2030 (which translates into a cumulative installed capacity of about 750 GW by 2030). On current trends, the EU is projected to reach only 718 GW by 2030, falling short of the 750 GW target. The Solar Power Europe report notes that the EU would benefit from a flexibility strategy to unlock the potential of battery storage and demand-side flexibility. According to [estimates](#) by the Commission's Joint Research Centre (JRC) Clean Energy Technology Observatory (CETO), installed capacity of solar thermal heat reached 41.7 GW thermal (GWth) in 2024. While the solar energy strategy recognises the importance of solar thermal technology, it does not set specific targets for its development.

The other main EU initiatives in the area of solar include the European [Solar PV Industry Alliance](#), launched in 2022, which aims to 'accelerate solar PV deployment in the EU by scaling-up to 30 GW of annual solar PV manufacturing capacity in Europe, facilitating investment, de-risking sector acceleration, and supporting Europe's decarbonisation targets'. The [European Solar Charter](#), signed in April 2024, sets out voluntary actions in support of the EU photovoltaic sector, such as promoting innovative forms of solar energy deployment and creating favourable framework conditions for manufacturing facilities for PV products.

Wind energy

According to a 2025 Wind Europe [report](#), total installed wind power capacity in the EU reached 236 GW (215 GW onshore and 21 GW offshore) in the first half of 2025. The 2020 [offshore renewable energy strategy](#) set non-binding targets of 60 GW of offshore wind and 1 GW of ocean energy by 2030, as well as 300 GW of offshore wind and 40 GW of ocean energy by 2050. These were then raised in the Member States' [pledge](#) (2023) to cumulative offshore wind and ocean energy of 111 GW by 2030 and of 317 GW by 2050. Wind Europe estimates that, to reach the 42.5 % renewables target, the EU would need to install 425 GW of cumulative offshore and onshore wind capacity. However, it projects, based on current trends, that the EU will achieve only 344 GW of installed wind capacity by 2030 (298 GW onshore and 46 GW offshore).

Additional EU initiatives include the [European wind power action plan](#) of October 2023, which sets out actions to accelerate permitting, improve auction systems and facilitate access to finance. The [European Wind Charter](#) of December 2023 includes voluntary commitments aimed at supporting the development of the EU wind sector.

Renewable hydrogen

According to the Clean Hydrogen Observatory, EU hydrogen consumption was [estimated](#) at 7.9 million tonnes in 2024. The 2025 CETO [report](#) on hydrogen indicates 7.2 million cubic metres of overall hydrogen production in 2024. The May 2022 [REPowerEU](#) plan set EU targets by 2030 of 10 million tonnes of domestic renewable hydrogen production and 10 million tonnes of imports. However, over [90 %](#) of hydrogen in the EU is produced with natural gas, involving significant GHG emissions. Water electrolysis, in combination with renewable electricity, is a key technology for the production of renewable hydrogen. According to the EU Agency for the Cooperation of Energy Regulators (ACER) 2025 European hydrogen markets monitoring [report](#), electrolyser capacity increased to 308 MW in 2024, which is significantly below the EU targets of 6 GW in 2024 and 40 GW in 2030 set out in the 2020 [EU hydrogen strategy](#). The strategy also established actions to boost investment, and demand for and scaling up of hydrogen production. Moreover, the [European Clean Hydrogen Alliance](#) was launched in 2020 to bring together stakeholders. The [Hydrogen Bank](#), established in 2022, is a financing instrument designed to improve investment opportunities. Furthermore, the EU recently adopted [rules](#) for renewable hydrogen production.

Bioenergy

Bioenergy, derived from agricultural, forestry and organic waste materials, made up [over half](#) of EU renewables consumption in 2023 (latest available data). Solid biofuels represented the largest share of bioenergy (68.1 %), followed by liquid biofuels (13.9 %), biogas and biomethane (11.2 %), and municipal waste (6.6 %). According to the [Regulation](#) on the Governance of the Energy Union and Climate Action (Article 35), the Commission must submit a report on bioenergy sustainability every two years. The 2003 RED revision strengthened the sustainability and GHG emissions saving [criteria for biomass](#), which were introduced in 2018. The directive also extended the 'no-go areas' for agricultural biomass to include forest biomass, and added new no-go area categories, such as primary and old growth forests, highly biodiverse forests and grassland, heathland, certain land with high carbon stock, and peatland. Biomass from these areas cannot be counted towards the renewables targets and receive support. Implementing regulations were adopted to provide uniform rules for the [implementation](#) of the sustainability criteria for forest biomass and on [certification](#) of forest biomass.

[Biofuels](#) are liquid transport fuels, such as biodiesel and bioethanol, made from biomass. The RED includes a combined sub-target for renewable hydrogen and advanced biofuels of 5.5 % in the transport sector by 2030. Since 2024, a [Union Database for Biofuels](#) (UDB) enables traceability of renewable and recycled carbon fuels.

The REPowerEU Plan set out a target of 35 billion cubic meters (bcm) annual biomethane production by 2030 (accompanied by several [measures](#) to boost it). The [Biomethane Industrial Partnership](#) (BIP), launched in 2022, seeks to support the achievement of the EU's 2030 target of 35 bcm of annual production and use of sustainable biomethane, and to create the conditions for a further ramp-up of its potential by 2050. The REPowerEU target for biogas use in power plants is 12.3 million tonnes of oil equivalent (Mtoe) by 2030.

Hydropower, geothermal and heat pumps

[Hydropower](#) is one of the oldest renewable energy sources and currently the second-largest renewable source of electricity after wind. It also plays an important role in electricity system flexibility and energy storage (in reservoirs). EU legislation aims to address the environmental impact of hydropower on wildlife and river morphology, mainly through the [Water Framework Directive](#) and environmental impact assessments. The EU also supports hydropower [research](#) and innovation projects.

Owing to technical and economic challenges, the development of [geothermal energy](#) has been slower in the EU compared with other energy sources. However, geothermal energy has an advantage over other sources, as it is not weather-dependent. While the net geothermal electricity capacity in the EU amounted to only 1 GW in 2022, there were 395 operating geothermal district heating systems. The 2016 EU [strategy](#) on heating and cooling highlighted the potential of geothermal energy and heat pumps in this sector. The [Heat Pump Accelerator Platform](#), launched in 2025, seeks to speed up the roll-out of heat pumps in buildings, industry and district heating. A new [EU heating and cooling strategy](#) is planned for the second quarter (Q2) of 2026. While the EU does not have a separate strategy on geothermal energy, several Member States have adopted national geothermal [roadmaps](#) with targets.

Investment in renewables

Investment in renewable energy in the EU has hit record highs. Annual average investment in energy transition (including renewables, grids and efficiency) reached around [€345 billion](#)³ in 2025, according to the International Energy Agency (IEA). Nevertheless, a significant investment gap remains to reach the legally binding targets for 2030 and 2050. According to the State of the Energy Union 2025 [report](#), the EU needs an average annual investment in the energy system of €660 billion from 2026 to 2030 to reach the 'fit for 55' targets (reducing GHG emissions by 55 % and achieving a 42.5 % renewable share in gross final energy consumption). This figure rises to €695 billion annually for 2031 to 2040, as the share of electricity in final energy consumption must increase from the current 23 % to approximately [50 %](#) in 2040. The European Investment Bank (EIB) [TechEU](#) programme is expected to mobilise €250 billion by 2027 to secure clean-tech supply chains. Furthermore, the EIB intends to deliver more than €75 billion of financing for 2026 to 2028 in support of the objectives of the energy transition as announced in the clean energy investment strategy.

Outlook

Renewables are set to continue their growth in Europe, largely driven by an expansion of solar and wind power. Over the 2005–2024 period, renewables' share in the EU final energy consumption more than doubled. More recently, renewables grew by 1 % in Europe year-on-year (2023–2024). Nevertheless, if the 2030 renewables target is to be met, the growth rate of renewables deployment needs to double, according to the [European Environment Agency](#) (EEA).

The development of renewables has so far been driven by dedicated policies and government support schemes (for instance, feed-in-tariffs), as well as increased economic competitiveness of renewables following a drastic fall in [generation costs](#).

Modelling by the [IEA](#) and [Ember](#) think tank shows that reaching the EU 2030 target is achievable if prompt action is taken to improve access to capital for investment in renewables and energy efficiency. The Commission's [guidance](#) on contracts for difference (CfDs), published in December 2025, addresses the issue of renewables project developers' access to capital by providing clarity on how governments can smartly design this crucial State aid tool without distorting price signals and maintaining the integrity of energy markets.

Strong growth of renewables is likely to reduce average electricity prices significantly. The EEA [estimates](#) that growth in renewables generation capacity in Europe could 'lower the variable costs of EU electricity generation by up to 57 % below 2023 levels'.

Trends per sector

Transport. The EU's trajectory for the uptake of renewable energy in transport is defined by the target of 29 % within the final energy consumption in the sector by 2030, or a 14.5 % GHG intensity reduction, as established in the 2023 RED III. The 2030 outlook rests on the 'fit for 55' framework, which prioritises the rapid electrification of passenger road transport and the scaling of sustainable aviation fuels and maritime e-fuels. While road transport should reach nearly 100 % electrification in new sales by 2035, according to [Regulation \(EU\) 2023/851](#) on emission performance standards,⁴ prominent think tanks such as [Agora Energiewende](#) and [Transport & Environment](#) (T&E) highlight that the 2050 climate-neutrality goal requires a deep transformation involving around [2 800 terawatt-hours \(TWh\)](#) of renewable electricity for the transport sector alone (for comparison, total EU electricity generation capacity in [2023](#) was 2 637 TWh). By 2050, the sector is expected to be powered by a mix, where direct electrification dominates road and rail transport, while renewable hydrogen and synthetic hydrocarbons provide the backbone for hard-to-abate heavy-duty trucking, shipping and aviation.

Current trends suggest that reaching the 2030 and 2050 targets remains difficult due to significant implementation gaps. The Commission's 2025 State of the Energy Union report indicates that transport remains the sector that is most resilient to decarbonisation efforts. While battery electric vehicle sales have surged, and charging infrastructure is currently growing faster than the car fleet itself, the EEA [warns](#) that actual transport emissions are only projected to fall roughly 4 % below 1990 levels by 2030 under current policies.

Looking towards 2050, Agora Energiewende [emphasises](#) that the EU transport sector is a major 'bottleneck' in the energy transition, projecting that, while the first 80 % of decarbonisation is technologically feasible through direct electrification, the final 20 % faces severe challenges, particularly in the aviation and maritime sectors.

Heating. The prospects for renewable energy in the EU heating sector are characterised by a transition towards direct electrification and high-efficiency district heating. The sector accounts for [nearly half](#) of the EU's final energy consumption. Under the RED III, the EU has set a binding target to increase the share of renewables in heating and cooling by at least 0.8 pp annually until 2025, and 1.1 pp from 2026 to 2030. According to the Commission, meeting the 2030 climate goals requires heat pump installations of approximately [30 million](#) units by the end of the decade. In 2024, some 2.31 million heat pumps were sold in 19 European countries, according to the European Heat Pump Association (EHPA).

The main obstacle to an accelerated uptake of heat pumps in Europe is the unfavourable price ratio between electricity and gas; this is because, in many Member States, high taxes and levies on electricity make heat pumps [more expensive](#) to operate than fossil fuel boilers, despite their superior efficiency. Additionally, the high upfront investment costs and a shortage of skilled installers create further hurdles. These issues are compounded by information gaps regarding subsidies, and a need for urgent power grid upgrades to handle the increased load from decentralised, electrified heating systems.

Main hurdles

Looking ahead, there are three main hurdles to accelerated deployment of renewables in the EU.

1. **Access to capital.** Renewables project developers face high capital costs, since the upfront investments in wind and solar generation facilities are substantial, while the operating costs are much [lower](#) due to the absence of fuel costs. Various tools, such as CfDs or feed-in-tariffs, can be used to address this challenge. The Commission put forward a [clean energy investment strategy](#) on 10 March 2026 to tackle some of these most pressing challenges.
2. **Electricity grids.** Electricity infrastructure in Europe needs to develop at a fast pace to accommodate the rise in renewables generation capacity. The estimated costs are substantial: the development of transmission and distribution grids would require [investment](#) of €1.2 trillion by 2040. In addition to physical expansion of the grids, substantial progress on this front can be made through the use of grid-enhancing technologies, digitalisation and flexible connection contracts. ⁵
3. **Permitting.** Permitting procedures for the installation of new renewable generation facilities and their connection to the grid are among the key bottlenecks to a faster roll-out of renewables in Europe. Permitting procedures are often fragmented across multiple layers of government, leading to administrative timelines that can take up to nine years for [wind](#) and four years for [solar](#) – far exceeding the two-year legal limit set under the RED III. These delays are driven by a lack of resources at national and local level, a lack of digitalised 'one-stop shop' portals, and inconsistent application of 'overriding public interest' status, which often leaves projects vulnerable to lengthy judicial appeals.

The Commission is assessing the current EU policy approach on renewables, and is looking at the post-2030 framework and the needs to be addressed. A legislative proposal on setting-up of the renewable energy framework is expected to be [published](#) in Q3 2026.

Main references

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Endnotes

- ¹ The most common statistical measure of the 'energy mix' is [gross available energy](#), which represents the quantity of energy necessary to satisfy all the energy demands. It is calculated according to the following formula: gross available energy = primary production + recovered and recycled products + imports – export + stock changes.
- ² EU targets for renewables set in the Renewable Energy Directive are expressed as the share of renewable energy in [gross final energy consumption](#). This covers the energy used by end-consumers (final energy consumption) plus grid losses and self-consumption by power plants. It does not include transformation losses and energy sector own-use, which are included in primary energy consumption (which is by definition larger than final energy consumption).
- ³ The figure quoted in the IEA report on World Energy Investment is expressed in USD (\$390 billion).
- ⁴ The regulation is currently under review, so the target may be adjusted accordingly.
- ⁵ See separate EPRS briefings on [TEN-E](#), on [acceleration of permit-granting procedures](#), and on [CfDs and grid connection](#) for more detail.

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Contact

E-mail: epers@ep.europa.eu

Intranet: <https://epers.in.ep.europa.eu/>

Internet: <https://www.europarl.europa.eu/thinktank>

Blog: <https://epthinktank.eu>